CareFlow AI — Interview Master Guide

Full Technical Instructions, Architecture, Technology Stack & Interview Preparation

Executive Summary: CareFlow AI is an enterprise-grade agentic healthcare orchestration platform designed for CAREPLUS Multispeciality Hospitals to automate and standardize inpatient discharge workflows. It incorporates 11 FastAPI microservices, stateful graph execution via **LangGraph**, Model Context Protocol (**MCP**) tool standards, Agent-to-Agent (**A2A**) protocol, **FAISS RAG** clinical knowledge retrieval, and mandatory **Human-in-the-Loop (HITL)** clinician approval.

1. Setup & Execution Instructions

Follow these exact commands to demonstrate or run the end-to-end platform locally:

Prerequisites: Python 3.11+ | Node.js v18+ & npm

```
# 1. Backend Service Execution (FastAPI + LangGraph + WebSockets)
cd 'CareFlow AI'
python -m pip install -r backend/requirements.txt
python -m uvicorn app.main:app --app-dir backend --reload --port 8000
```

- **Backend REST API Docs:** http://localhost:8000/docs
- **WebSocket Telemetry Stream:** ws://localhost:8000/ws/telemetry

```
# 2. Frontend Enterprise Console Execution (React + Vite + Tailwind)
cd frontend
npm install
npm run dev
```

- **Frontend UI Application:** http://localhost:3000

```
# 3. Automated Test Suite Execution
python -m pytest backend/tests
```

2. Technology Stack & Key Frameworks

Component	Technology Used	Key Features & Role
Frontend UI	React 18, TypeScript, Vite	High-performance SPA with modern HMR and strict type safety.
Styling & Icons	Tailwind CSS, Lucide Icons	Glassmorphic dark theme, responsive enterprise grid layout.
Backend Framework	FastAPI, Uvicorn	Asynchronous REST API, auto OpenAPI generation, high throughput.
Real-Time Engine	WebSockets (ConnectionManager)	Bi-directional streaming for live agent states & telemetry.
Workflow Engine	LangGraph	Stateful Directed Acyclic Graphs (DAG) with checkpoints & HITL.
Tool Standard	Model Context Protocol (MCP)	Standardized schema registration for 11 microservice tools.
Agent Protocol	A2A Protocol	Multi-agent collaboration across 6 specialized AI roles.
Vector Database	FAISS + NumPy	Fast similarity search over clinical policies & guidelines.
Observability	LangFuse-style Tracing	Execution metrics, token usage tracking, & latency waterfalls.

3. System Architecture & 11 Microservices

The platform is structured into **11 Microservices** orchestrated seamlessly by LangGraph:

- **1. Patient Service:** Ingests demographic and EHR history data.
- **2. Clinical Summary Service:** Aggregates lab results and clinical summaries.
- **3. Medication Reconciliation Service:** Scans for drug-drug interactions (e.g., Warfarin + Aspirin).
- **4. Discharge Planning Service:** Calculates readiness scores and criteria.
- **5. Document Generation Service:** Compiles multi-lingual discharge summaries using LLMs.
- **6. Follow-up Service:** Schedules post-discharge appointments and rehabilitation plans.
- **7. Insurance / TPA Service:** Checks coverage with Star Health API & Vidal TPA.
- **8. Pharmacy Service:** Verifies outpatient medication inventory.
- **9. Notification Service:** Formats WhatsApp/SMS notifications for caregivers.
- **10. Risk & Safety Service:** Evaluates 30-day readmission risk scores.
- **11. Audit & Telemetry Service:** Records immutable logs for compliance.

4. Comprehensive Interview Preparation (Q&A;)

Q1: Can you give a high-level overview of CareFlow AI?

CareFlow AI is an agentic hospital discharge orchestration platform. It automates patient discharge processing using 11 FastAPI microservices and 6 autonomous AI agents coordinated by LangGraph. It ensures clinical safety through drug interaction checks, FAISS RAG policy verification, and mandatory Human-in-the-Loop clinician sign-off before discharge instructions are finalized.

Q2: Why did you choose LangGraph for agent orchestration?

Hospital discharge workflows require state persistence, conditional branching, and deterministic guardrails—which traditional linear LLM chains cannot handle. LangGraph allows us to express the workflow as a stateful Directed Acyclic Graph (DAG). If high readmission risk or drug conflicts are detected, conditional edges dynamically route execution to extra audit nodes. Additionally, LangGraph provides native support for state checkpoints, enabling Human-in-the-Loop approval.

Q3: What is MCP (Model Context Protocol) and how is it implemented?

MCP is an open protocol standard that decouples tools from specific LLM frameworks. In CareFlow AI, each microservice exposes standardized MCP tools (such as get_patient, check_medication_conflicts, and check_insurance_tpa). This allows any agent engine (LangGraph, Google ADK, or Agno) to interact with our microservices without custom wrapper code.

Q4: How do you enforce medical safety and compliance?

Medical safety is enforced at three distinct levels: 1) Deterministic rule engines that cross-reference drug interactions; 2) FAISS RAG clinical policy verification; and 3) A strict Human-in-the-Loop (HITL) checkpoint that blocks document dispatch until a licensed clinician reviews and signs off on the plan in the UI.

Q5: How does real-time observability work in CareFlow AI?

We implemented a custom telemetry system inspired by LangFuse. Every agent tool invocation logs model details (gemini-3.6-flash), prompt version, latency waterfall, and token counts. These audit records are saved to an immutable telemetry store and streamed live to the frontend Observability Dashboard via WebSockets.

5. Key Resume / Portfolio Impact Bullet Points

- **Architected CareFlow AI:** Enterprise agentic discharge platform featuring 11 FastAPI microservices and 6 autonomous agents.
- **Stateful Workflow DAGs:** Utilized **LangGraph** for state persistence, conditional risk-based routing, and Human-in-the-Loop approval checkpoints.
- **Interoperability Standard:** Implemented **Model Context Protocol (MCP)** for standardized microservice tool access.
- **Real-Time Observability:** Developed real-time telemetry streaming via **WebSockets** with token usage and latency waterfall analysis.